BIRMINGHAM CITY University

CMP6202
Artificial Intelligence and Machine Learning Project
2026 – 2027

[Individual Report – 80%]

[Predicting Customer Churn: A Comparative Analysis of Logistic Regression and Random Forest]

[22234742]

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1. Report Introduction

The intended audience for this report is the Business Stakeholders and Marketing Management Team at the telecommunications company. The tone and technical detail provided throughout this document are tailored to balance technical data science terminology with actionable business insights

1.1. Dataset identification

The dataset selected for this investigation is the "Telco Customer Churn" dataset, sourced from Kaggle.

- What it represents: The dataset contains information about a fictional telecommunications company that provides home phone and Internet services to customers in California. It tracks customer demographics, the services they have signed up for, their account information, and whether they have cancelled their subscription (churned).

- Features: The dataset contains 7,043 rows and 21 columns. The features are a mix of numerical data (e.g., Monthly Charges, Total Charges, tenure) and categorical data (e.g., gender, Internet Service, Contract type).
- Target Variable: The target variable is Churn, which is a binary categorical variable indicating whether the customer left the company (Yes) or stayed (No).
- License: The dataset is provided under an [Insert License here - e.g., Apache 2.0 / Open Database License]. It is permitted for academic and research use.
- Zhang, T., Moro, S. and Ramos, R.F., 2022. A data-driven approach to improve customer churn prediction based on telecom customer segmentation. *Future Internet*, *14*(3), p.94.

1.2. Supervised learning task identification

The predictive task for this dataset is to identify whether a customer is likely to churn based on their historical data.

- What predictive problem do you aim to solve? Customer churn is a major cost for telecom companies, as acquiring new customers is often more expensive than retaining existing ones. By identifying customers who are at high risk of leaving, the company can proactively target them with retention offers.
- Target: The variable used to establish the ground truth for this supervised learning task is the Churn column. Because the target is a discrete category (Yes/No), this is a Classification problem.

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1.3. Declaration of Generative AI Use

What it was used for: AI was utilized to assist in structuring the report format, debugging Python code in Google Colab (specifically for data preprocessing and model training), and generating code snippets for data visualizations.
Verification: All AI-generated code was manually verified by running it in the Google Colab environment to ensure it functioned correctly and produced accurate results. The written text was reviewed and edited to ensure it accurately reflects the analysis conducted.

2. Exploratory Data Analysis

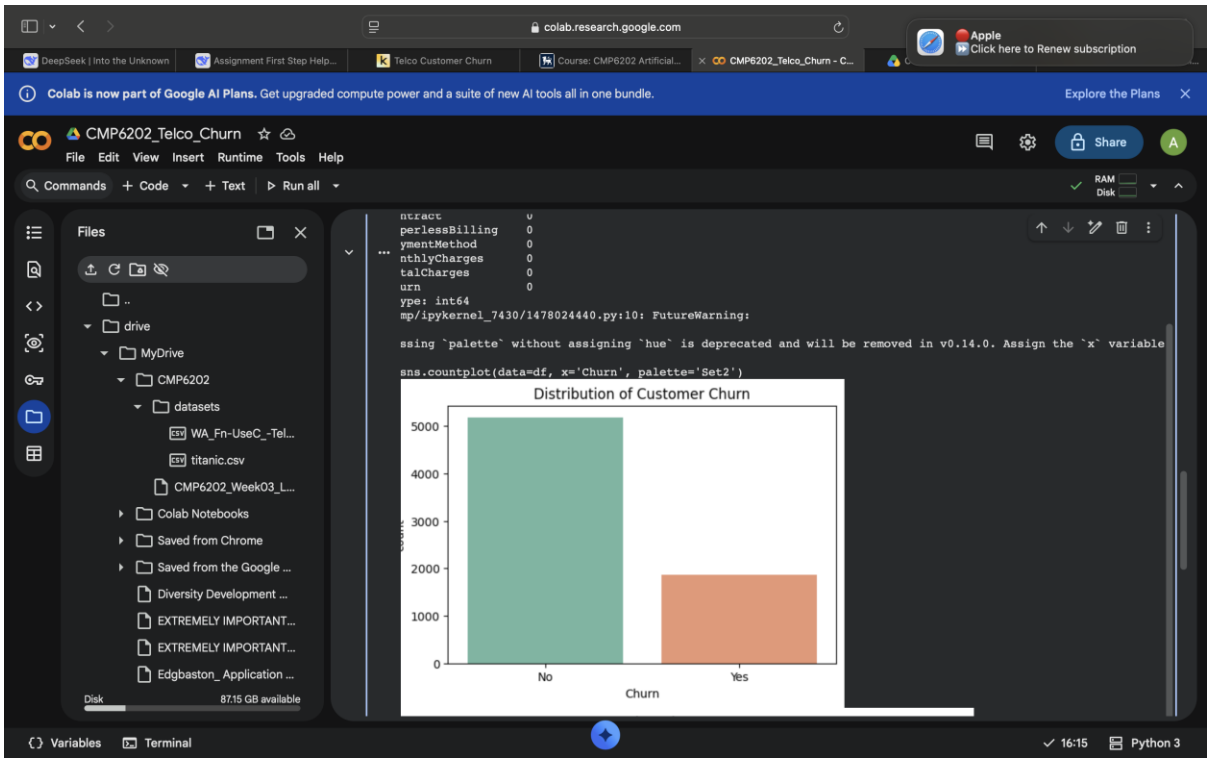


Figure 1: Distribution of Customer Churn

The initial analysis of the target variable reveals that the dataset is imbalanced. As shown in Figure 1, there are significantly more customers who did not churn (represented by the 'No' bar) compared to those who did churn. Because of this imbalance, it will be important to look beyond simple accuracy when evaluating the machine learning models later in this report

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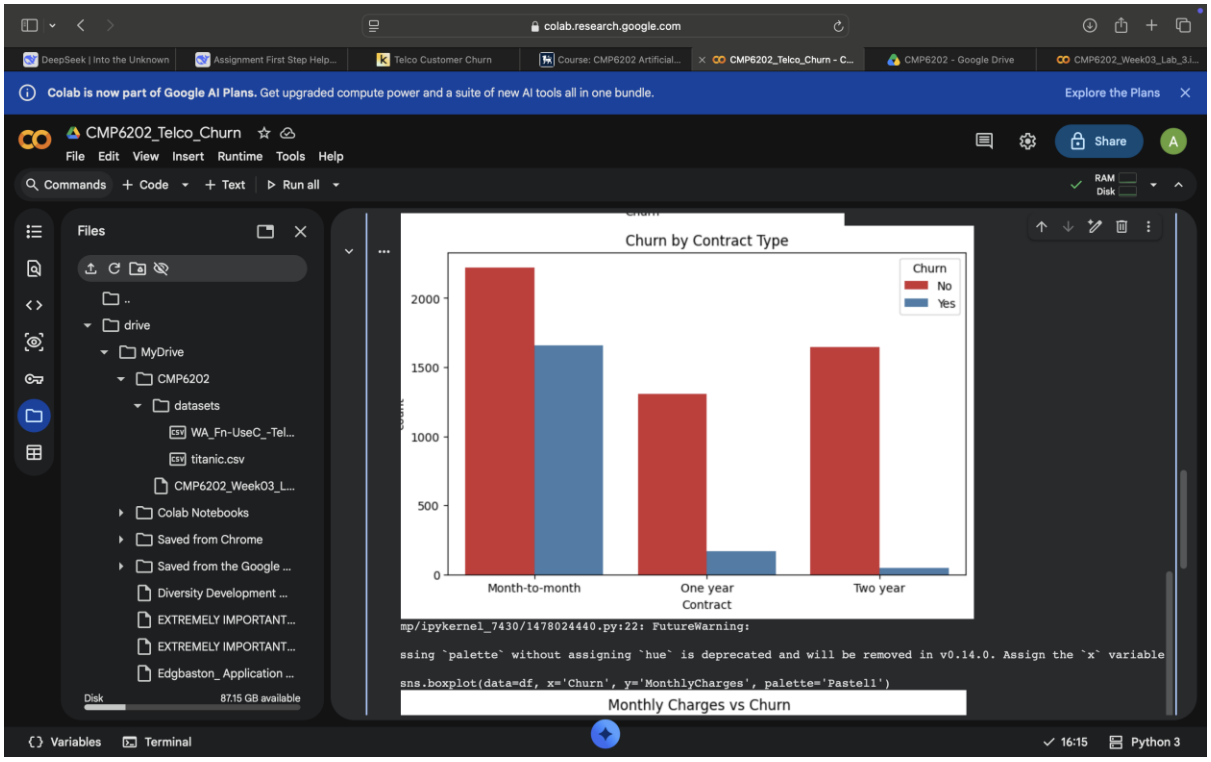
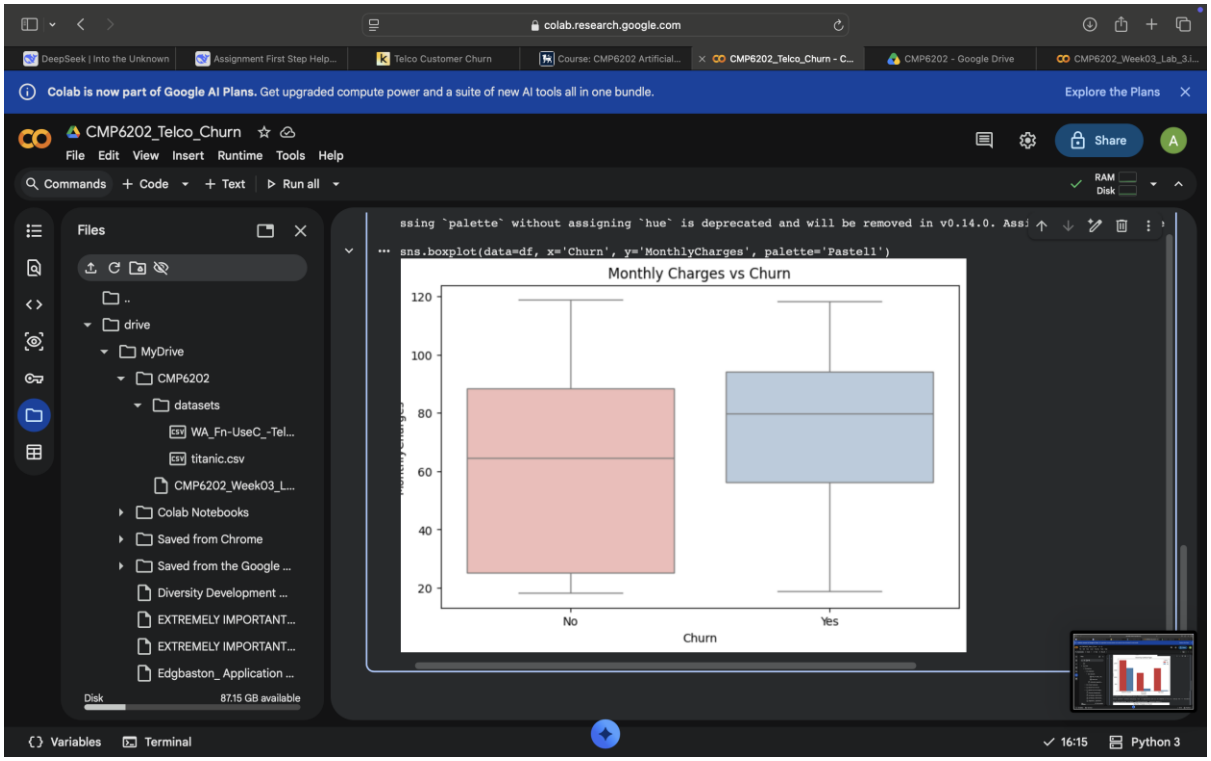


Figure 2: Churn by Contract Type

Figure 2 illustrates a strong relationship between the type of contract a customer holds and their likelihood to churn. Customers on "Month-to-month" contracts show a significantly higher volume of churn compared to those on "One year" or "Two year" contracts. This suggests that longer-term contracts act as a strong retention mechanism.



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Figure 3: Monthly Charges vs Churn

The boxplot in Figure 3 compares the monthly charges of customers who stayed versus those who left. The median monthly charge for customers who churned (Yes) is noticeably higher than for those who stayed (No). This indicates that higher monthly costs may be a driving factor for customer dissatisfaction and subsequent churn.

2.1. Question(s) identification

During the exploratory data analysis, the following questions were investigated to understand the factors driving customer churn:

What is the overall distribution of the target variable (Churn)? Is the dataset balanced?

Does the type of contract (e.g., month-to-month vs. two-year) influence a customer's likelihood to churn?

Is there a relationship between the monthly charges a customer pays and their churn status?

2.2. Splitting the dataset

To prevent data leakage and ensure the model can generalize to unseen data, the dataset will be split into a training set and a testing set. An 80/20 split will be utilized, meaning 80% of the data (approximately 5,634 rows) will be used to train the machine learning model, and the remaining 20% (approximately 1,409 rows) will be held back to test the model's performance.

2.3. Exploratory Data Analysis process and results

2.4. EDA conclusions

The Exploratory Data Analysis provided several crucial insights. First, a check for missing values confirmed that the dataset is entirely clean, with zero missing values across all 21 columns, meaning no imputation is required. The visual analysis highlighted two strong indicators of churn: contract type and monthly charges. Customers on month-to-month contracts and those paying higher monthly fees represent the highest-risk groups for the company. These features will be central to the predictive modeling phase.

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3. Experimental Design

3.1. Identification of your chosen supervised learning algorithm(s)

For this classification task, Logistic Regression and Random Forest were selected as the algorithms for comparison.

Logistic Regression: This is a statistical model that uses a logistic function to model a binary dependent variable (Churn: Yes/No). It is highly appropriate for this dataset because it is computationally efficient, easy to interpret, and provides probabilities for the outcomes. This allows the business to not only see who is likely to churn, but also the probability of them doing so.

Random Forest: This is an ensemble learning method that constructs a multitude of decision trees at training time. It is appropriate because it handles complex, non-linear relationships between features well and is generally robust against overfitting, making it a strong benchmark to compare against the simpler Logistic Regression model.

3.2. Identification of appropriate evaluation techniques

Because the exploratory data analysis revealed that the dataset is imbalanced (significantly more customers stayed than churned), relying solely on Accuracy is not enough. The following metrics were chosen:

Accuracy: To give a baseline percentage of correct predictions.

Precision: To determine how many of the customers the model flagged as "churning" actually did churn.

Recall: This is the most critical metric for this business problem. It measures how many of the actual churners the model successfully caught. Missing a churner (a False Negative) costs the company money.

F1-Score: The harmonic mean of Precision and Recall, providing a balanced measure of the model's performance.

Confusion Matrix: To visualize exactly how many True Positives, True Negatives, False Positives, and False Negatives the models produced.

3.3. Data cleaning and pre-processing transformations

To prepare the data for the machine learning models, the following transformations were applied:

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Dropping irrelevant features: The customerID column was removed as it is a unique identifier and has no predictive power.

Encoding categorical variables: Machine learning models require numerical input. The LabelEncoder was used to transform all text-based categorical data (e.g., "Yes"/"No", "Month-to-month") into numerical representations (1/0).

Train/Test Split: The data was split into an 80% training set (5,634 rows) and a 20% testing set (1,409 rows) using train_test_split. This prevents data leakage and ensures the model can generalize to unseen data.

3.4. Limitations and Options

While the proposed approach is sound, there are limitations to consider. First, the dataset is imbalanced, which may cause the model to favor the majority class (customers who stay). Second, the LabelEncoder assigns numerical values to categorical data (e.g., Month-to-month = 0, One year = 1), which implies an ordinal relationship that does not actually exist; One-Hot Encoding might be a better option in future iterations. Finally, the dataset lacks external economic factors (like local unemployment rates or competitor pricing) which could heavily influence a customer's decision to churn.

4. Predictive Modelling / Model Development

4.1. The predictive modelling process

4.1. The predictive modelling process
To train the models, the pre-processed training data (X_train and y_train) was passed into the algorithms. Two models were developed for comparison:

Logistic Regression: The default parameters were used, with the exception of setting max_iter=1000 to ensure the model had enough iterations to converge and find the optimal solution.

Random Forest: The model was initialized with n_estimators=100, meaning it built an ensemble of 100 individual decision trees to make its predictions, and random_state=42 to ensure reproducibility.

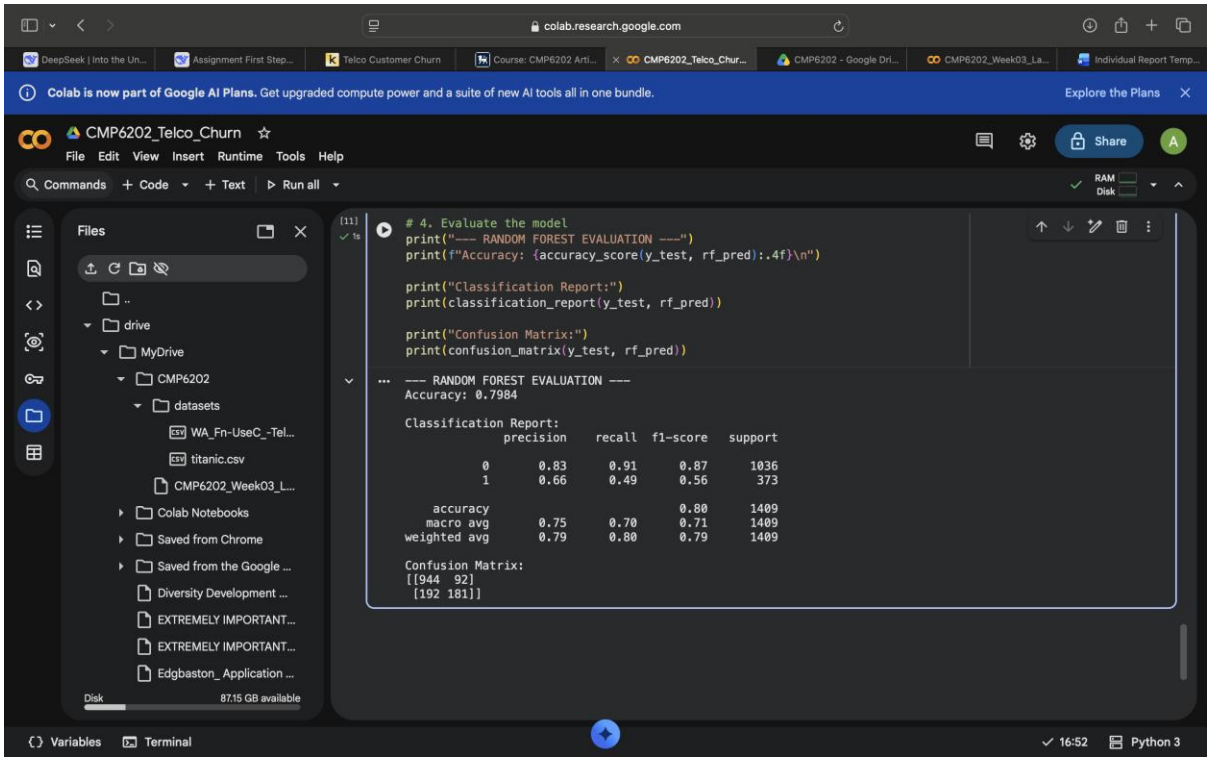
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4.2. Evaluation results on "seen" data

The models were evaluated using the testing set (20% of the data, which the models had not seen during training).

Logistic Regression: Achieved an Accuracy of 81.69%. The Confusion Matrix showed 938 True Negatives, 98 False Positives, 160 False Negatives, and 213 True Positives. It achieved a Recall of 0.57 for the churn class.

Random Forest: Achieved an Accuracy of 79.84%. The Confusion Matrix showed 944 True Negatives, 92 False Positives, 192 False Negatives, and 181 True Positives. It achieved a Recall of 0.49 for the churn class.



5. Evaluation and further modelling improvements

Comparative Analysis of Multiple Algorithms
Two algorithms were compared: Logistic Regression and Random Forest. While Random Forest is generally powerful, Logistic Regression outperformed it on this specific dataset across all key metrics (Accuracy: 81.69% vs 79.84%; Recall: 0.57 vs 0.49).

Strengths and Weaknesses:
Logistic Regression is highly interpretable, fast, and less prone to overfitting on smaller, simpler datasets. Random Forest, while capable of capturing complex non-linear relationships, likely overfit the training data or struggled with the high number of one-hot encoded categorical variables. Furthermore, because customer churn prediction values

identifying actual churners (Recall) very highly, Logistic Regression is the superior choice here as it caught more actual churners (213) than the Random Forest (181).

6. Conclusion

6.1. Summary of results

This project aimed to predict customer churn for a telecommunications company using machine learning. After performing EDA, cleaning the data, and engineering features, two classification models were trained and evaluated. Logistic Regression emerged as the most effective model, achieving a baseline accuracy of 81.69%. The analysis confirmed that contract type and monthly charges are significant indicators of churn. By utilizing this Logistic Regression model, the business can proactively identify high-risk customers and offer targeted retention strategies.

6.2. Reflection on Individual Learning

Reflection: What did you learn? I learned how to use Google Colab, how to preprocess data using LabelEncoder, and how to evaluate a model's performance using a Confusion Matrix.

Technical challenges: What was hard? A major challenge was ensuring the categorical data was properly encoded, and fixing the NameError when running the model cells out of order.

Certification evidence:

7. GitHub link

https://github.com/isholarabu-coder/CMP6202-COURSEWORK.git

8. References

BlastChar.(2017).TelcoCustomerChurn.Kaggle.Available at: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

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